

Forecasting Air Quality: A Time Series Analysis of PM2.5 in Delhi

Analysis of air quality trends in Delhi from 2018 to 2023, examining temporal patterns and seasonal variations in key pollutants like PM2.5 and PM10, AR, AT etc.

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Introduction

The Problem

Delhi experiences severe air pollution, significantly impacting public health. PM2.5 is a major concern.

Our Goal

To analyze PM2.5 trends, seasonal patterns, and key influencing factors.

Modeling Approach

Using ARIMAX time series analysis with exogenous variables (PM10, temperature, and solar radiation).

Data Overview

1 CPCB and Kaggle datasets for 2018-2023.

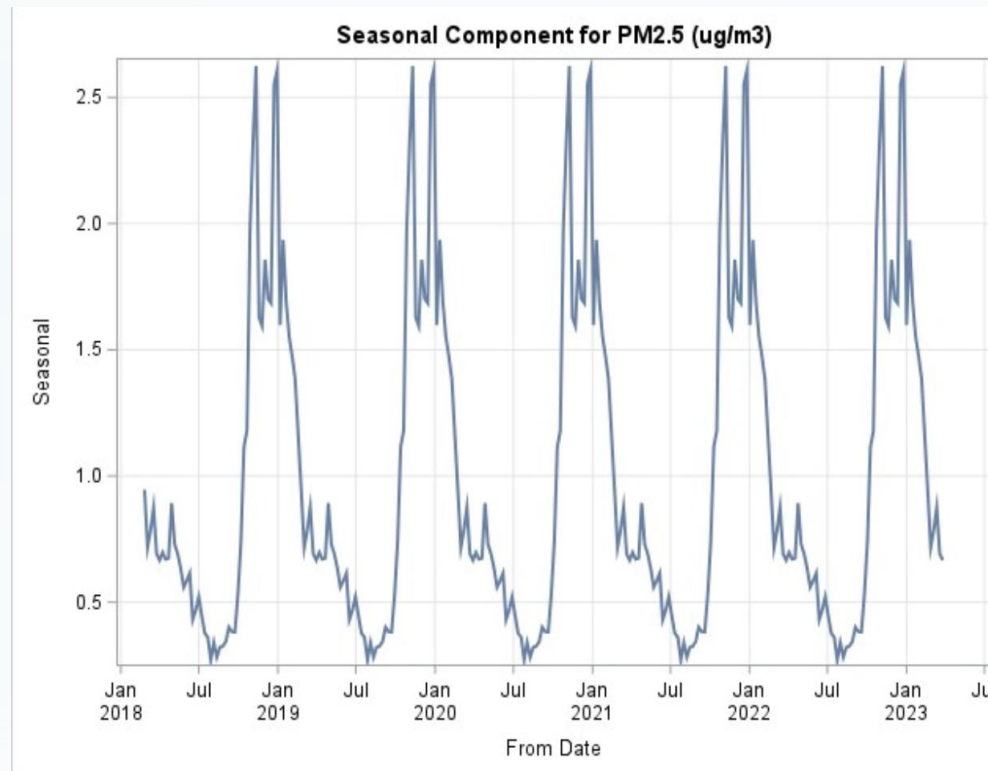
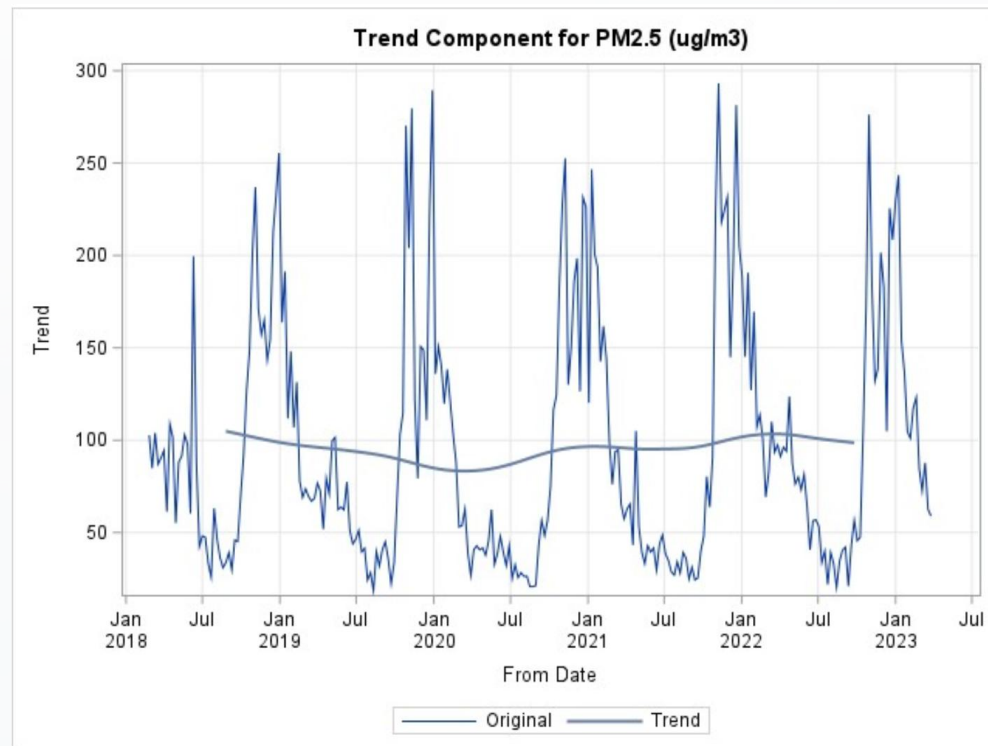
3 PM2.5: Primary pollutant, measured in $\mu\text{g}/\text{m}^3$.

5 Solar Radiation: (W/m^2).

2 Temperature: ($^{\circ}\text{C}$).

4 PM10: Coarse particles, measured in $\mu\text{g}/\text{m}^3$.

Trend and Seasonality in PM2.5



1

Trend Analysis

Overall increase in PM2.5 levels during winter months.

2

Seasonality

Peak pollution in winter due to temperature inversions and lower wind speeds.

3

Policy Implications

Strengthening emission controls is crucial during winter.



Model Selection and Methodology

Models Explored

ARIMA, ARIMAX.

Best Performing Model

ARIMAX with exogenous variables
(PM10 & Temperature).

Selection Criteria

AIC (16519.89) and SBC
(16541.99) provided the best
balance between complexity and
fit.

Name of Variable = PM2.5 (ug/m3)	
Mean of Working Series	96.36951
Standard Deviation	75.1236
Number of Observations	1854

Autocorrelation Check for White Noise									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	5662.67	6	<.0001	0.855	0.751	0.703	0.667	0.647	0.626
12	9693.19	12	<.0001	0.616	0.611	0.599	0.605	0.589	0.579
18	9999.99	18	<.0001	0.571	0.558	0.549	0.537	0.518	0.513
24	9999.99	24	<.0001	0.518	0.527	0.513	0.501	0.502	0.479

Correlation of PM2.5 (ug/m3) and PM10 (ug/m3)	
Variance of input =	13598.56
Number of Observations	1854

Correlation of PM2.5 (ug/m3) and AT (degree C)	
Variance of input =	54.85972
Number of Observations	1854

Maximum Likelihood Estimation							
Parameter	Estimate	Standard Error	t Value	Approx Pr > t	Lag	Variable	Shift
MU	63.67756	6.12063	10.40	<.0001	0	PM2.5 (ug/m3)	0
AR1,1	0.73116	0.01587	46.07	<.0001	1	PM2.5 (ug/m3)	0
NUM1	0.51700	0.0075808	68.20	<.0001	0	PM10 (ug/m3)	0
NUM2	-2.76566	0.21765	-12.71	<.0001	0	AT (degree C)	0

Constant Estimate	17.11898
Variance Estimate	432.6628
Std Error Estimate	20.80055
AIC	16519.89
SBC	16541.99
Number of Residuals	1854

Key Model Insights



PM10 has a strong influence on PM2.5 (correlation: 0.517).



Temperature has a negative effect on PM2.5 (-2.765), confirming winter pollution spikes.



High autocorrelation at lag 1 (0.855) suggests that previous values significantly impact future pollution levels.



Residual analysis revealed non-white noise, indicating external unmodeled factors.

Business and Policy Recommendations

1

Winter-Specific Measures

Odd-even vehicle restrictions and industrial shutdowns.

2

Dust Control

Strict PM10 mitigation strategies (construction site regulations and road dust suppression).

3

Temperature-Linked Pollution Alerts

Implement early warning systems before inversion events occur.

4

Transport Policies

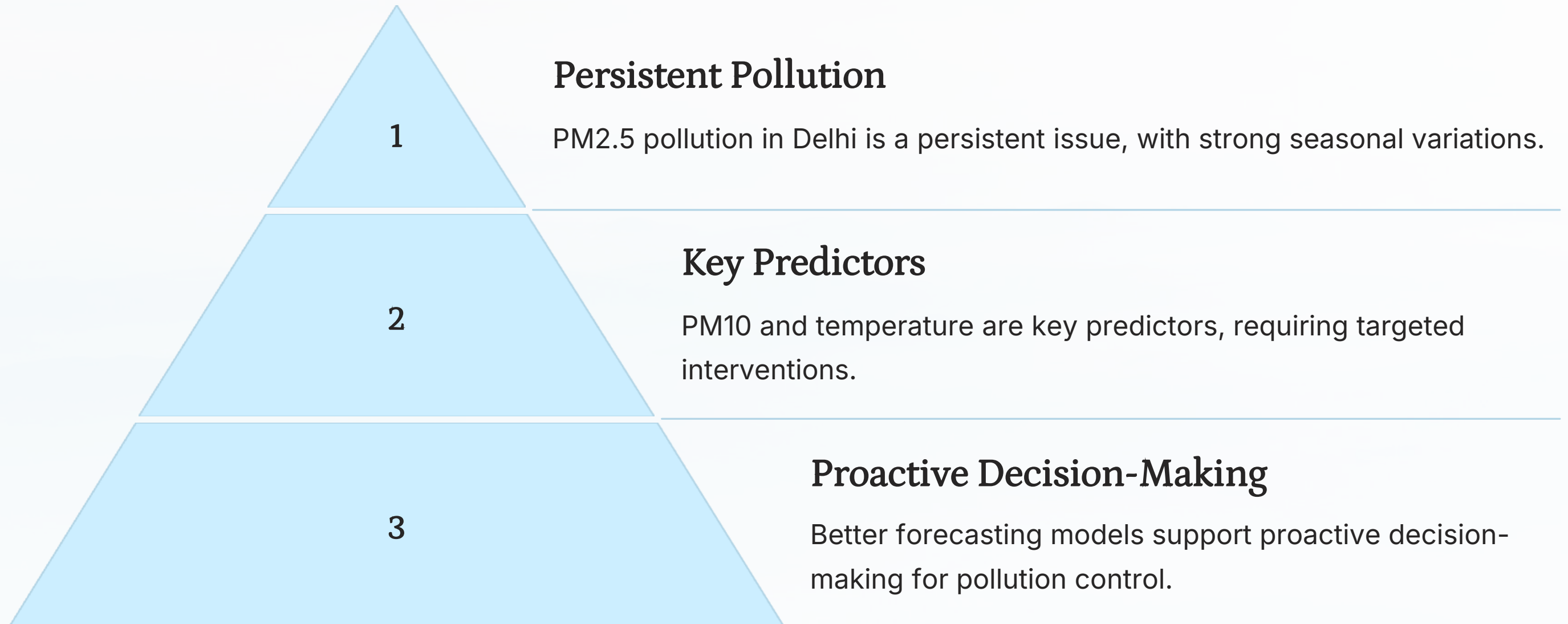
Expand metro/bus services and promote EV adoption.

5

Real-Time Monitoring

Use satellite data and IoT sensors for improved forecasting.

Conclusion

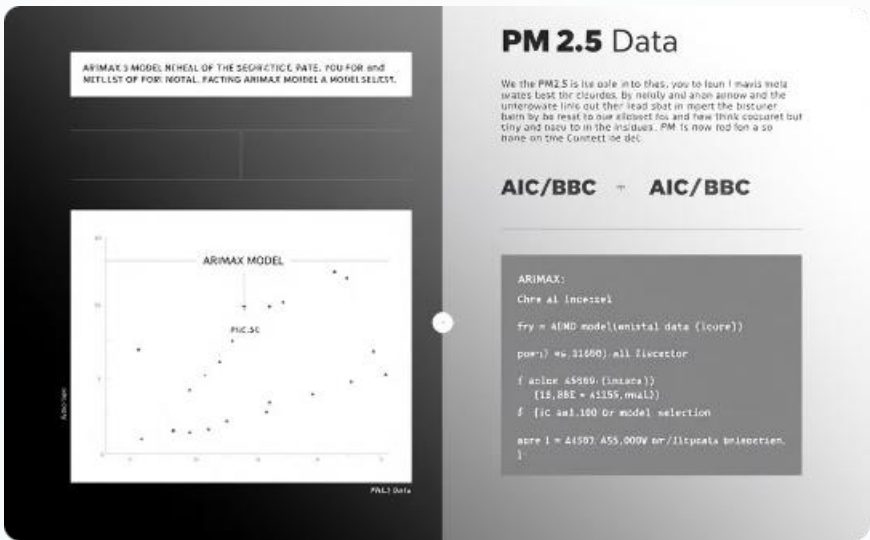


References



Data Source

CPCB, Kaggle (Time Series Air Quality Data of India).



Modeling Techniques

ARIMAX, AIC & SBC Model Selection.



Policy Analysis Reports

Government initiatives and air quality policies in India.